

Technical Interview Question Bank

1. Python & Programming Fundamentals

Basic

1. What is a variable in Python?
2. What is the difference between a list and a tuple?
3. What is a dictionary?
4. What is a function?
5. What is a class?
6. What is a loop?
7. What is an exception?
8. What is a module?
9. What is the difference between == and is?
10. Why is Python widely used in AI/ML?

Intermediate

1. List vs tuple — when would you use each?
2. What is list comprehension?
3. What is a lambda function?
4. What is a Python generator?
5. What is the difference between shallow copy and deep copy?
6. What is a virtual environment?
7. Why are NumPy arrays preferred over Python lists for numerical operations?
8. What is broadcasting in NumPy?
9. How would you handle missing values using Python?
10. How would you identify a slow part of a Python program?

Advanced

1. Why can Python loops be slow for numerical operations?
2. What is the GIL and how can it affect Python applications?

3. Threads vs processes for CPU-intensive Python workloads?
4. How would you reduce memory consumption in a Python ML pipeline?
5. How would you optimize a slow Pandas operation?
6. How would you make an ML codebase modular and maintainable?
7. How would you make an ML experiment reproducible?
8. How would you manage conflicting Python dependencies?
9. How would you design an efficient Python inference service?
10. How would you profile and optimize a memory-heavy ML application?

2. Data Structures & Algorithms

Basic

1. What is a data structure?
2. What is an array/list?
3. What is a stack?
4. What is a queue?
5. What is a hash map?
6. What is a tree?
7. What is a graph?
8. What is searching?
9. What is sorting?
10. What is Big-O notation?

Intermediate

1. When can binary search be used?
2. Why is hash-map lookup generally fast?
3. What is BFS?
4. What is DFS?
5. What is a heap?

6. What is a sliding window?
7. What is dynamic programming?
8. How would you find duplicate elements efficiently?
9. How would you find the most frequent element?
10. Why is algorithm complexity important when processing ML datasets?

Advanced

1. How would you find nearest neighbors efficiently?
2. Why does high dimensionality make nearest-neighbor search difficult?
3. How would you process millions of graph relationships?
4. How would you reduce an $O(n^2)$ preprocessing operation?
5. How would you design a streaming frequency counter?
6. What is approximate nearest-neighbor search?
7. How would you process a dataset larger than available RAM?
8. How would you choose an algorithm for a large-scale feature pipeline?
9. How would you identify an algorithmic bottleneck?
10. How would you verify that an optimized algorithm still produces correct results?

3. Mathematics & Statistics for ML

Basic

1. What is mean?
2. What is median?
3. What is mode?
4. What is variance?
5. What is standard deviation?
6. What is probability?
7. What is a vector?
8. What is a matrix?

9. What is correlation?
10. Why is statistics important in machine learning?

Intermediate

1. Mean vs median — which is better when outliers exist?
2. What does standard deviation tell you?
3. What is conditional probability?
4. What is Bayes' theorem?
5. What is covariance?
6. Correlation vs causation?
7. What is feature normalization?
8. What is a dot product?
9. Why are vectors important in machine learning?
10. What is a probability distribution?

Advanced

1. Why does feature scaling affect some ML algorithms?
2. What is multicollinearity?
3. Why can correlation be misleading?
4. What is the bias-variance trade-off?
5. What is a confidence interval?
6. What does a p-value tell you?
7. Why does high dimensionality create problems?
8. What are eigenvectors used for in machine learning?
9. Why are logarithms commonly used in probability and ML?
10. How would you detect whether the distribution of production data has changed?

4. Data Handling & SQL

Basic

1. What is a dataset?
2. What is a feature?
3. What is a label?
4. What is missing data?
5. What is an outlier?
6. What is SQL?
7. What is a primary key?
8. What is a JOIN?
9. What is filtering?
10. Why is data cleaning important?

Intermediate

1. How would you handle missing values?
2. How would you identify outliers?
3. What is feature encoding?
4. Normalization vs standardization?
5. How would you find duplicate records using SQL?
6. Why are database indexes useful?
7. What is data leakage?
8. Why should preprocessing be fitted only on training data?
9. How would you process a very large CSV file?
10. What is an ETL pipeline?

Advanced

1. How would you prevent data leakage in an ML pipeline?
2. How would you process terabytes of data?
3. How would you design a feature store?

4. What is training-serving skew?
5. How would you optimize a slow analytical SQL query?
6. When would denormalization be useful for ML systems?
7. How would you handle changing data schemas?
8. How would you validate a data pipeline?
9. How would you make feature generation reproducible?
10. How would you monitor data quality in production?

5. Machine Learning Fundamentals

Basic

1. What is machine learning?
2. Supervised vs unsupervised learning?
3. What is a feature?
4. What is a label?
5. What is training?
6. What is testing?
7. What is overfitting?
8. What is underfitting?
9. What is classification?
10. What is regression?

Intermediate

1. Why do we split data into training and testing sets?
2. What is validation data?
3. What is cross-validation?
4. Why is regularization used?
5. What is feature scaling?
6. What is class imbalance?

7. Why should we create a baseline model?
8. What is hyperparameter tuning?
9. What is a loss function?
10. What is a model parameter?

Advanced

1. Why can training accuracy be misleading?
2. How would you detect overfitting?
3. How does regularization improve generalization?
4. How would you handle severe class imbalance?
5. What is data leakage?
6. How would you select an ML algorithm for a new problem?
7. What is model calibration?
8. Why use stratified cross-validation?
9. What is concept drift?
10. How would you decide whether an ML model is ready for production?

6. ML Algorithms & Model Evaluation

Basic

1. What is linear regression?
2. What is logistic regression?
3. What is a decision tree?
4. What is clustering?
5. What is accuracy?
6. What is precision?
7. What is recall?
8. What is F1 score?
9. What is a confusion matrix?

10. What is K-means clustering?

Intermediate

1. When is accuracy a poor evaluation metric?
2. Precision vs recall?
3. When would you use ROC-AUC?
4. What is cross-validation used for?
5. Decision tree vs linear regression?
6. What is ensemble learning?
7. What is random forest?
8. What is gradient boosting?
9. Why do some ML algorithms require feature scaling?
10. How would you select a classification threshold?

Advanced

1. Why can ROC-AUC hide poor minority-class performance?
2. When is PR-AUC more useful?
3. How would you tune a model without data leakage?
4. Calibration vs discrimination?
5. Why can boosting models overfit?
6. How would you compare two ML models fairly?
7. How would you optimize a model for low latency?
8. Why does feature importance not necessarily imply causation?
9. How would you evaluate a model under distribution shift?
10. How would you select the operating threshold for a real-world ML system?

7. Deep Learning

Basic

1. What is a neural network?
2. What is a neuron?
3. What is an activation function?
4. What is an epoch?
5. What is a batch?
6. What is a loss function?
7. What is backpropagation?
8. What is a CNN?
9. What is an embedding?
10. What is dropout?

Intermediate

1. Why is ReLU commonly used?
2. What is the vanishing-gradient problem?
3. Why do we normalize input data?
4. CNN vs fully connected network?
5. What is transfer learning?
6. What is batch normalization?
7. What is learning rate?
8. What is an optimizer?
9. Why do we need validation data?
10. What is data augmentation?

Advanced

1. Why can deep neural networks overfit?
2. How would you diagnose vanishing gradients?
3. How would you reduce the size of a deep-learning model?

4. What is attention?
5. Why are Transformers effective for sequence tasks?
6. What is fine-tuning?
7. What is catastrophic forgetting?
8. How would you optimize model inference latency?
9. How would you prevent training-serving preprocessing mismatch?
10. How would you evaluate a deep-learning model beyond accuracy?

8. Generative AI & LLMs

Basic

1. What is an LLM?
2. What is a token?
3. What is prompting?
4. What is an embedding?
5. What is hallucination?
6. What is RAG?
7. What is fine-tuning?
8. What is a context window?
9. What is temperature in LLM generation?
10. What is an LLM API?

Intermediate

1. Why use RAG instead of only prompting?
2. How does semantic search work?
3. What is chunking in a RAG system?
4. Why can a RAG system still hallucinate?
5. What is prompt injection?
6. How would you evaluate an LLM application?

7. Fine-tuning vs RAG?
8. What is an embedding model used for?
9. Why should retrieved information be traceable to sources?
10. How would you reduce LLM token costs?

Advanced

1. How would you design a production RAG system?
2. How would you reduce hallucinations?
3. How would you defend against prompt injection?
4. How would you evaluate retrieval quality?
5. What is reranking?
6. How would you handle sensitive data in an LLM application?
7. How would you monitor an LLM application's quality?
8. How would you choose between a small and large language model?
9. How would you reduce incorrect citations generated by an LLM?
10. How would you design an AI agent with controlled tool access?

9. MLOps & Model Deployment

Basic

1. What is model deployment?
2. What is inference?
3. What is a model artifact?
4. What is an API?
5. What is a container?
6. What is model monitoring?
7. What is latency?
8. What is batch inference?
9. What is real-time inference?

10. Why is versioning important in ML?

Intermediate

1. Batch inference vs real-time inference?
2. What should be versioned in an ML project?
3. How would you expose an ML model through an API?
4. Why use containers for ML deployment?
5. What is CI/CD for ML?
6. What should model monitoring measure?
7. What is data drift?
8. What is concept drift?
9. How would you roll back a bad model?
10. Why should training and inference preprocessing be consistent?

Advanced

1. How would you design an end-to-end ML deployment pipeline?
2. How would you detect model drift?
3. How would you handle training-serving skew?
4. How would you deploy a large model efficiently?
5. What is canary deployment for ML models?
6. How would you ensure ML experiment reproducibility?
7. How would you monitor a model when labels are delayed?
8. How would you control inference costs?
9. How would you define automated rollback criteria for a model?
10. How would you secure a production ML API?

10. AI/ML Project Technical Depth

Basic

1. What problem does your AI/ML project solve?
2. What dataset did you use?
3. Which model did you use?
4. Why did you choose that model?
5. What features did you use?
6. What preprocessing did you perform?
7. How did you split your dataset?
8. Which evaluation metric did you use?
9. What part of the project did you personally implement?
10. What was the biggest technical challenge?

Intermediate

1. Why did you select that particular algorithm?
2. How did you handle missing data?
3. How did you prevent data leakage?
4. How did you tune your model?
5. How did you evaluate generalization?
6. What baseline model did you compare against?
7. What happens if production data differs from training data?
8. How is your model integrated into the application?
9. What is the weakest technical component of your project?
10. What would you technically improve if you had another month?

Advanced

1. How would you scale your AI solution to production?
2. How would you detect model failure in production?
3. How would you reduce inference latency?

4. How would you improve model robustness?
5. How would you handle severe distribution shift?
6. How would you explain your model's predictions?
7. How would you secure your AI/data pipeline?
8. How would you fairly compare two different model architectures?
9. What technical assumption in your project is most likely to fail?
10. If you rebuilt the project from scratch, what technical architecture would you change and why?